

Visual features

A useful feature responds to something relevant in an image while staying stable under changes that should not alter the answer.

15–20 MINUTES

Prerequisite: image tensors

BROWSER COMPARISON

1 • Signal

Color, edge, texture, or shape.

2 • Filter

A local rule produces a response.

3 • Invariance

Ignore irrelevant variation.

4 • Failure

A shortcut breaks outside training.

From raw values to evidence

Raw pixels are sensitive. Move an object one pixel, change the lamp, or alter exposure and many values change. A **feature** is a representation that makes some pattern easier to use: a color range, an edge, a corner, a texture, or a learned combination.

A classical filter is an explicit rule. For example, an edge filter compares nearby intensities. A strong change produces a large response. A color threshold marks pixels inside a chosen numeric range. Learned systems discover many filters from data rather than having a programmer specify each one.

Central mental model

A feature should change when task-relevant evidence changes and remain stable when irrelevant conditions change.

Local filters

A convolution-like image filter slides a small grid of weights across the image. At each location it multiplies neighboring values by the weights and sums them. The result is a **feature map**: a new grid describing where that pattern appears.

$$\text{response} = \sum (\text{local pixel} \times \text{filter weight})$$

An edge kernel with negative weights on one side and positive weights on the other responds to a transition. It does not “understand an object”; it detects a local pattern that later processing can combine.

Robust cues and shortcuts

Color is relevant to our class names, but it is not automatically reliable. A red surface in shadow may produce different values from the same surface under bright light. A red sticker may share color but not shape. A detector must use evidence that distinguishes cubes from other colored objects.

A **shortcut** is a correlation that works in the training data but does not express the intended task. If every blue cube is photographed on one background, a model may use the background. The training score can improve while the useful behavior does not.

Browser exercise: test feature robustness

Rule A

“Select every strongly red pixel.”

Rule B

“Combine color, closed edges, corners, and context.”

Imagine a red ball, a red poster, a dark red cube, and a partly hidden cube. Predict which rule produces more false positives and which might miss the shadowed cube.

Reveal the analysis

Applied to our cubes

GENERAL PRINCIPLE Color thresholding is a useful baseline and diagnostic: it can reveal camera or lighting problems. It does not by itself enforce cube shape.

REPOSITORY DECISION The project requires shape discrimination and uses a learned object detector as its main path. Vendor color-detection examples remain references rather than the production recognizer.

HYPOTHESIS When the detector behaves differently in the robot room, lighting, scale, background, viewpoint, and camera processing are candidate causes. Each needs a controlled test.

Project anchor: project definition.

Check your understanding

Which description best captures a robust visual feature?

- ☐ Stable under irrelevant changes
- ☐ Identical for every image
- ☐ Largest raw pixel value

Explain it back

Explain why “redness” is useful evidence for a red cube but is not a complete detector. Give one relevant variation and one irrelevant variation for this task.

Primary sources

- Szeliski — image processing, filtering, edges, and recognition.
- D2L: Convolutions for Images — local filters and feature maps.